

Problem Understanding

The task is to analyze lexical tokens generated during compilation. Since raw token counts do not provide meaningful insights into code complexity, a derived feature called *Token Density* is used. It measures the number of tokens per line in the source code.

Logic Used

The program takes two inputs: total number of tokens and total number of lines.

It computes Token Density using the formula:

$$\text{Token Density} = \text{Tokens} / \text{Lines}$$

A threshold value of 5.0 is used to classify the code:

- If Token Density > 5.0 → Token Heavy Code
- Otherwise → Normal Code

Justification of Threshold

The threshold value of 5.0 is chosen because:

- A higher number of tokens per line indicates dense and complex code
- Such code reduces readability and maintainability
- Lower density indicates well-structured and readable code